1. Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

2. Load Dataset

```
url = "https://raw.githubusercontent.com/nunnarilabs/ml/master/house/kc_hous

df = pd.read_csv(url)

df.head()
```

	id	date	price	bedrooms	bathrooms	sqft_living	s
0	7129300520	20141013T000000	221900.0	3	1.00	1180	
1	6414100192	20141209T000000	538000.0	3	2.25	2570	
2	5631500400	20150225T000000	180000.0	2	1.00	770	
3	2487200875	20141209T000000	604000.0	4	3.00	1960	
4	1954400510	20150218T000000	510000.0	3	2.00	1680	

5 rows × 21 columns

3. Basic Data Exploration

```
df.info()
df.describe()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21613 entries, 0 to 21612
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   id             21613 non-null  int64
 1   date           21613 non-null  object
 2   price          21613 non-null  float64
 3   bedrooms       21613 non-null  int64
 4   bathrooms      21613 non-null  float64
 5   sqft_living    21613 non-null  int64
 6   sqft_lot       21613 non-null  int64
 7   floors         21613 non-null  float64
 8   waterfront     21613 non-null  int64
 9   view           21613 non-null  int64
```

4. Data Cleaning

```
# Drop unnecessary columns if needed
df = df.dropna()

# Verify again
df.isnull().sum()
```

```
 17  lat            21613 non-null  float64
 18  long           21613 non-null  float64
 19  sqft_living15  21613 non-null  int64
 20  sqft_lot15     21613 non-null  int64
dtypes: float64(5), int64(15), object(1)
memory usage: 3.5+ MB
```

	0
id	0
date	0
price	0
bedrooms	0
bathrooms	0
sqft_living	0
sqft_lot	0
floors	0
waterfront	0
view	0
condition	0
grade	0
sqft_above	0
sqft_basement	0
yr_built	0
yr_renovated	0
zipcode	0

lat	0
id	0
date	0
price	0
bedrooms	0
bathrooms	0
sqft_living	0
sqft_lot	0
floors	0
waterfront	0
view	0
condition	0
grade	0
sqft_above	0
sqft_basement	0
yr_built	0
yr_renovated	0
zipcode	0
lat	0
long	0
sqft_living15	0
sqft_lot15	0

dtype: int64

5. Exploratory Data Analysis (EDA)

```
corr_matrix = df.corr(numeric_only=True)

corr_matrix
```

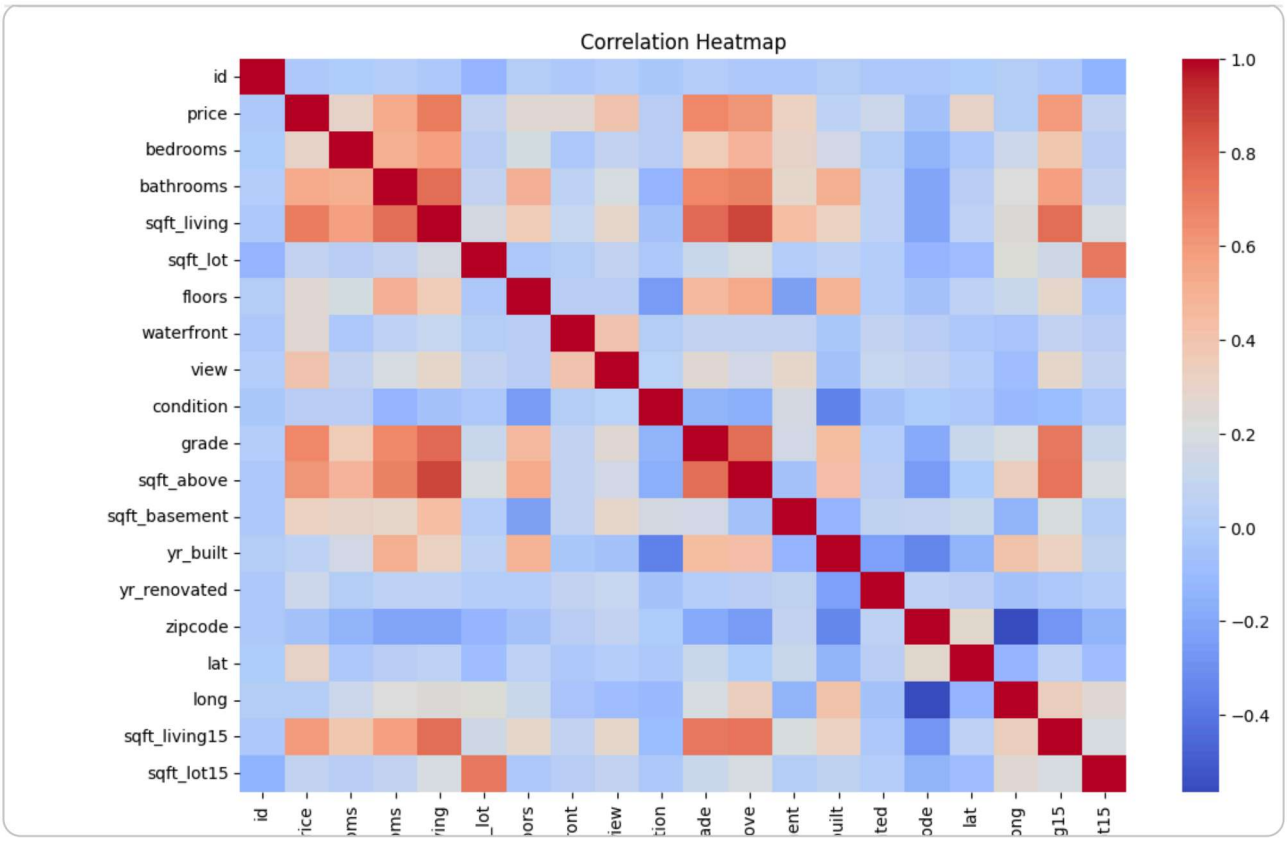
	id	price	bedrooms	bathrooms	sqft_living	sqft_lo
id	1.000000	-0.016762	0.001286	0.005160	-0.012258	-0.13210
price	-0.016762	1.000000	0.308350	0.525138	0.702035	0.08966
bedrooms	0.001286	0.308350	1.000000	0.515884	0.576671	0.03170
bathrooms	0.005160	0.525138	0.515884	1.000000	0.754665	0.08774
sqft_living	-0.012258	0.702035	0.576671	0.754665	1.000000	0.17282
sqft_lot	-0.132109	0.089661	0.031703	0.087740	0.172826	1.00000
floors	0.018525	0.256794	0.175429	0.500653	0.353949	-0.00520
waterfront	-0.002721	0.266369	-0.006582	0.063744	0.103818	0.02160
view	0.011592	0.397293	0.079532	0.187737	0.284611	0.07471
condition	-0.023783	0.036362	0.028472	-0.124982	-0.058753	-0.00895
grade	0.008130	0.667434	0.356967	0.664983	0.762704	0.11362
sqft_above	-0.010842	0.605567	0.477600	0.685342	0.876597	0.18351
sqft_basement	-0.005151	0.323816	0.303093	0.283770	0.435043	0.01528
yr_built	0.021380	0.054012	0.154178	0.506019	0.318049	0.05308
yr_renovated	-0.016907	0.126434	0.018841	0.050739	0.055363	0.00764
zipcode	-0.008224	-0.053203	-0.152668	-0.203866	-0.199430	-0.12957
lat	-0.001891	0.307003	-0.008931	0.024573	0.052529	-0.08568
long	0.020799	0.021626	0.129473	0.223042	0.240223	0.22952
sqft_living15	-0.002901	0.585379	0.391638	0.568634	0.756420	0.14460
sqft_lot15	-0.138798	0.082447	0.029244	0.087175	0.183286	0.71855

Next steps: Generate code with `corr_matrix` New interactive sheet

6. Data Visualization

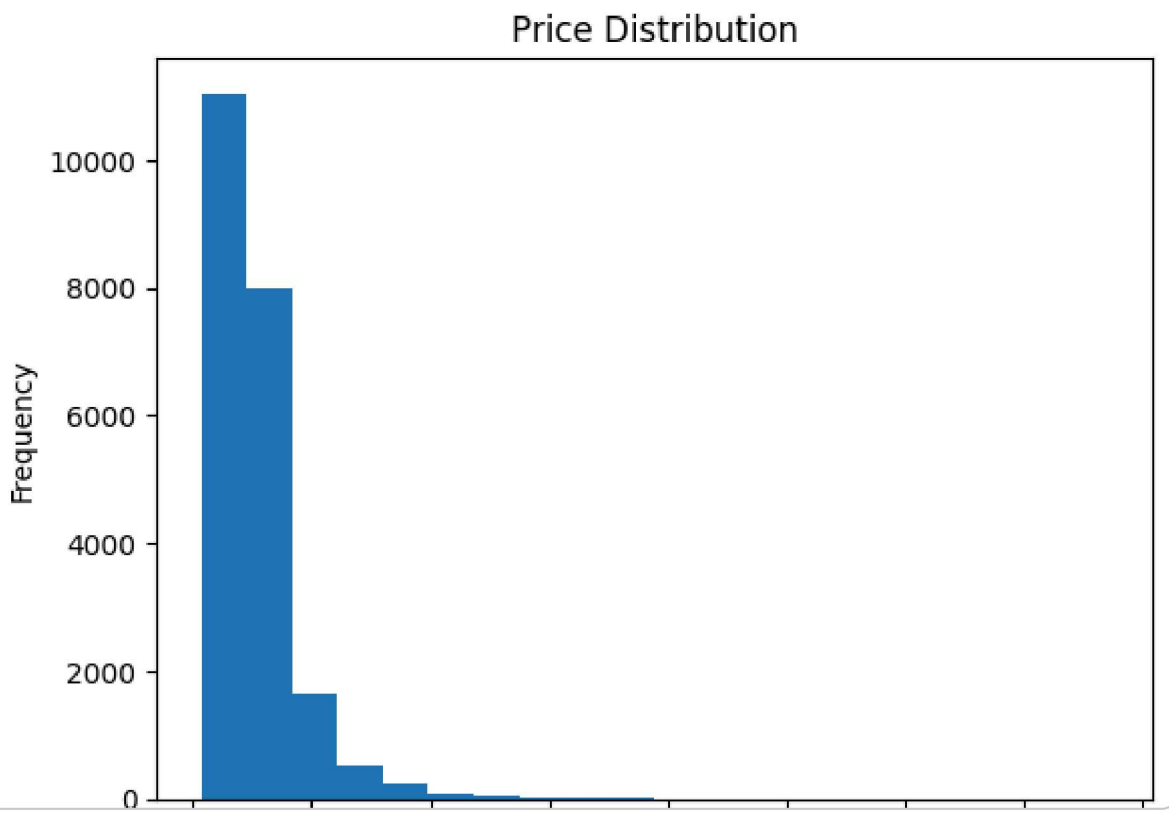
Heatmap

```
plt.figure(figsize=(12,8))
sns.heatmap(corr_matrix, annot=False, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



Histogram

```
plt.hist(df['price'], bins=20)
plt.title("Price Distribution")
plt.xlabel("Price")
plt.ylabel("Frequency")
plt.show()
```



Scatter Plot

```
plt.scatter(df['sqft_living'], df['price'])
plt.xlabel("Living Area (sqft)")
plt.ylabel("Price")
plt.title("Living Area vs Price")
plt.show()
```

Seaborn